

Description of the ascii geo-VLBI data format AGVF

Introduction

Ascii geo-VLBI data format (AGVF) is designed for support of transferring VLBI Level-2 data between software packages and their editing. Since AGVF is in plain ascii, a development of a handler for reading/writing is a trivial task and does not require specialized skills of installation of specialized software. The AGVF was derived from Mark-3 DBH format used in 1976–2018. A handler for AGVF format was implemented in 2006 and incorporated into Solve software package. The library gvh that implements the handler is distributed also as a standalone package.

Since AGVF format is ascii, its parsing incurs certain overheads. It is not recommended for adoption as an internal format for data analysis software packages. The main use case of AGVF format is a distribution of Level-2 data and results of data analysis of a specific VLBI experiment for geodesy and absolute astrometry between various data analysis software packages. Software analysis packages are supposed to be able to transform from the AGVF format to their internal format and vice versa without loss of information.

Level 2 VLBI data in the context of this document means results of fringe fitting program in a form of total group delay, total phase delay rate, and the total phase. The datafile in AGVF format contains also other variables that describe the VLBI experiment and results of its analysis. In the context of this document a set of data that corresponds to what was scheduled independently is called **experiment**. Experiment duration varies from 30 minutes to 15 days, although durations less than 1 hour and longer than 24 hours are unusual. During an experiment N stations observe M sources. Duration of time when an array or its part observe a given source before slewing to another source is called **scan**. Data collected by a pair or stations during scan is called **observation**.

Structure of an AGVF file

AGVF file consists of records of variable length. Records consist of ascii symbols with codes 32 through 255 (decimal). Characters with codes 128 to 255 are allowed but discouraged. A record consists of words separated by one or more blanks. Records have a prefix in a form of a 4-character long section ID followed by character . (dot) followed by the chunk index.

A valid AGVF file consists of the format label and one or more chunks. Each chunk consists of 5 mandatory sections in the following order:

- **FILE** section
- **PREA** (preamble) section
- **TOCS** (table of contents) section
- **DATA** section
- **HEAP** heap section. Always empty. Reserved for future use.

Sections **FILE**, **PREA**, and **TOCS** describe the data, Section **DATA** presents the data. The data are considered as a set of four-dimensional arrays of one of the types: CHARACTER*1, INTEGER*2, INTEGER*4, INTEGER*8, REAL*4, REAL*8 and of one of the classes: **SES**, **SCA**, **STA**, **BAS**. Fortran notation is used for describing arrays: a) dimensions start from 1; b) the index over first dimension runs first, i.e. elements order is a(i,j), a(i+1,j), a(i+2,j) etc. The dimensions 3 and 4 depend on the class of the data.

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Throughout this document the strings that should be present in the AGVF file are **highlighted**. Examples that are not a part of definitions are also **highlighted**.

For historical reasons an individual array of data is called LCODE. An LCODE defines a 8-character long name, a string of arbitrary length with its description, class, type, and four dimensions. A class determines the scope of the LCODE and dimensions of the high level. Supported classes:

- **SES** The scope of the LCODE is the entire experiment, f.e. instance the experiment name. The dimensions 3 and 4 are 1, 1.
- **SCA** The scope of the LCODE is a given scan. The third dimension is the number of scans, the fourth dimension is 1.
- **STA** The scope of the LCODE is a given station, for instance apriori clock delay used for correlation. The third dimension is the number of observations at a given station, the fourth dimension is the station index.
- **BAS** The scope of the LCODE is a given observation, f.e. group delay. The third dimension is the total number of observations, the fourth dimension is 1.

An LCODE can be defined only once. There are five mandatory LCODEs that should be defined at the beginning of the first chunk. The AGVF format does not regulate which other LCODEs should be present, in which order, and in which chunk. The order of codes and their allocation in the chunk is selected by the generating software.

AGVF label

The AGVF label a string of 64 characters long the identifies the format and its revision. The AGVF label is the first record of a valid AGVF file.

Example:

AGV format of 2005.01.14

File section

A record in the file section has two words. The first word is the record prefix **FILE** and the chunk index separated by dot. Chunks are numbered starting from 1 consecutively. The second word is the the full path file name that contributed to the chunk.

Example:

FILE.1 /vlbi/gvf/db/20170117_pr1775_fr1_v001.bgv

Preamble section

Preamble section contains auxiliary information. A record of the preamble sections has format Prefix Keyword Value Unit separated a blank. Prefix has format: **PREA** and the chunk index separated by dot. Unit can be empty. A preamble section of the first chunk has three mandatory keywords:

- **@section_length**: defines the number of records in the preamble section except this one. This record should be the first record of the section. The first word of the Value is the number of records, the second is unit: keyword.
- **GENERATOR**: defines name and version of the program that generated this section. Version number

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should be defined that way that alphabetic sorting the value alphabetically should correspond to sorting the value chronologically.

- **CREATED_AT**: defines the creation date in format YYYY.MM.DD-hh:mm:ss

Example:

PREA.1 @section_length: 15 keywords

A preamble section of chunks others than the first has only one mandatory keyword: @section_length. An empty preamble section must have value 0 of this keyword.

Text section

Text section consists of auxiliary information, for instance, correlation report, schedule file, etc. Non-ascii text, such as document in pdf-format can be put in the text section after [base64](#) encoding. Text section consists of one or more chapters. A chapter is a portion of text. A record of the text sections has format Prefix Keyword Value. A keyword is allowed to be empty.

The first record of the text section defines the number of chapters. This is a mandatory record. Format: Prefix Keyword Value Unit.

- **Prefix** — **TEXT** and the chunk index separated by dot.
- **Keyword** — **@section_length**:
- **Value** — the number of chapters in the section as an integer number.
- **Unit** — **chapters**.

Example:

TEXT.1 @section_length: 1 chapters

The first record of a chapter has 8 mandatory fields followed by the chapter title. Chapter title can be empty. Format: Prefix Keyword1 Value1 Value2 Unit1 Keyword2 Value3 Unit2 [Title]

- **Prefix** — **TEXT** and the chunk index separated by dot.
- **Keyword1** — **@@chapter**
- **Value1** — the chapter index as an integer number. The first chapter has index 1.
- **Value2** — the number of records in this chapter as an integer number. the number of records is zero for an empty chapter
- **Unit1** — **records**
- **Keyword1** — **max_len**:
- **Value2** — the maximum number of characters in any record of this chapter.
- **Unit1** — **characters**.

Example:

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@@chapter 1 1 records, max_len: 16

The following n records have format: **TEXT** and chunk index separated by dot, one blank and the character string.

Table of contents section

The table of contents section defines LCODE names, their class, type, dimension and provides a brief LCODE description. The first record of the TOCS section defines the number of LCODEs defined in this chunk. Format of the first record: Prefix Keyword Value Unit.

- **Prefix** — **TOCS** and the chunk index separated by dot.
- **Keyword** — **@section_length:**
- **Value** — the number of chapters in the section as an integer number.
- **Unit** — **lcodes**.

Example:

TOCS.1 @section_length: 53 lcodes

The following records have format: Prefix LCODE Class Type Dim1 Dim2 Description

- **Prefix** — **TOCS** and the chunk index separated by dot.
- **Keyword** — LCODE -- a 8-characters long string with LCODE name. LCODE name should not contain blanks inside.
- **Class** — class of the LCODE. One of
 - **SES** — session class with the scope of entire experiment;
 - **SCA** — scan class with the scope of a given scan;
 - **STA** — station class with the scope of a given station;
 - **BAS** — baseline class with the scope of a given observation.
- **Type** — type of the data. One of
 - **C1** — Character*1 data. Each element is one byte;
 - **I2** — Integer*2, or 16-bit signed number;
 - **I4** — Integer*4, or 32-bit signed number;
 - **I8** — Integer*8, or 64-bit signed number;
 - **R4** — Real*4, or 32-bit real number;
 - **R8** — Real*8, or double precision, or 64-bit real number.
- **Dim1** — the first dimension of the data. 1 for scalar data.
- **Dim2** — the second dimension of the data. 1 for scalar or one-dimensional data.

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- **Description** — a brief text that describes the LCODE.

Example:

TOCS.1 SITNAMES SES C1 8 10 IVS site names

In this example LCODE **SITNAMES** is of session type (i.e. with the scope of entire experiment). It has character type and has dimensions 8, 10, 1, 1. The first dimension **8** defines the length of a string. The second dimension **10** defines the number of strings.

There are 5 mandatory keywords that must be defined at the beginning of the first chunk. They all have session class and integer*4 type.

1. **NUMB_OBS** — the total number of observations;
2. **NUMB_STA** — the total number of stations;
3. **NUMB_SCA** — the total number of scans;
4. **NOBS_STA** — a one-dimension array sized as NUMB_STA. Contains the number of observations per stations;
5. **OBS_TAB** — a two dimensional array sized as (3,NUMB_OBS). The first row of the array is the number of scans, the second row is the index of the first station at a baseline for this observation; and the third row is the index of the second station at a baseline for this observation.

Data section

One record of the data section keeps one element of the LCODE array.

The first record of DATA section defines the number of Data records present in this chunk. Format of the first record: Prefix Keyword Value Unit.

- **Prefix** — **DATA** and the chunk index separated by dot.
- **Keyword** — **@section_length:**
- **Value** — the number of chapters in the section as an integer number.
- **Unit** — **records**.

Example:

DATA.5 @section_length: 650859 records

The following records have format: Prefix LCODE Dim3 Dim4 Dim1 Dim2 Value

- **Prefix** — **DATA** and the chunk index separated by dot.
- **Lcode** — LCODE name.
- **Dim3** — the third dimension of the LCODE. The third dimension can be **0** or **1** for data of **SES** class.
- **Dim4** — the fourth dimension of the LCODE. The fourth dimension can be **0** or **1** for data of **STA** or **SES** class.
- **Dim1** — the first dimension of the LCODE.

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- **Dim2** — the second dimension of the LCODE.
- **Value** — the value of the array element. Recommended format for float numbers: 1PE15.7 for Real*4 data and for 1PD22.15 Real*8 data.

Example:

DATA.1 GR_DELAY 4466 0 1 1 7.267257847095946D-03

DATA.1 GR_DELAY 4466 0 2 1 7.267232471203255D-03

In this example group delay for observation with index 4466 is shown. The group delay was defined as array with dimensions 2,1,4538,1 in the table of contents. The first dimension runs over bands. The first value group delay corresponds to the 1st band and the second to 2nd band. The frequency of bands are determined in another LCODE (REF_FREQ in this case). The AGVF itself does not specify the unit of LCODE. Units definition is supposed to be defined in the LCODE description or in an external document.

HEAP section

Reserved for future use.

CHUN record

The last record of a data chunk contains the number of records in the chunk, excluding this record. The first line with the AGVF format label is counted. Format of the CHUN record: Prefix Keyword Value Unit:

- **Prefix** — **CHUN** and the chunk index separated by dot.
- **Keyword** — **@chunk_length:**
- **Value** — the number records in the chunk, excluding this record and counting the AGVF format label if this is the first chunk.
- **Unit** — **records.**

Example:

CHUN.4 @chunk_size: 72856 records

Comments on using the AGVF format

The AGVF format provides sufficient flexibility to carry results of fringe fitting, such as estimates of group delay, delay rate, their formal uncertainties, etc (Level-2A) and results of geodetic preprocessing, such as flags, solution parameterization, correction to weights, etc (Level-2B). AGVF is not suitable to carry Level-1 data (raw results of correlation).

Splitting an AGVF file into chunks is for facilitating an interface with other formats for Level-2 geodetic VLBI results. A valid AGVF file is supposed to have at least one chunk. Splitting data in chunks is determined by the generator process.

Some data are station dependent, for instance cable calibration, some are observation-dependent, for instance group delays. They are stored differently. Array OBS_TAB stored at the LCODE with the same name determines relationship between observation index and the indices of station-dependent parameters. For an observation with index **k** station-dependent LCODEs will have indices OBS_TAB(2,k) for the 1st (i.e. reference) station of a baseline and OBS_TAB(3,k) for the 2nd (i.e. remote) station of a baseline.

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History of LCODE naming is traced in 70s. Renaming LCODEs is discouraged because of a risk of breaking compatibility. Addition of new LCODEs carries a low risk.

Appendix

As an example, LCODEs for the experiment r1775 processed by PIMA are shown below. Only first five LCODEs are mandatory since are used by the handler for processing other LCODEs. LCODEs are grouped into two categories: common that originate from post-correlator analysis software or log-file parsing software that generates a Level-2A dataset and the data analysis specific LCODEs. The LCODEs in the latter category is mainly from geodetic/astrometry data analysis software packages that processes a Level-2A dataset and generates a Level-2B dataset (edited, reweighted, with parameterization stored).

Common LCODES

TOCS.1	NUMB_OBS	SES	I4	1	1	Number of observations in the session
TOCS.1	NUMB_SCA	SES	I4	1	1	Number of scans in the session
TOCS.1	NUMB_STA	SES	I4	1	1	Number of sites
TOCS.1	NOBS_STA	SES	I4	10	1	Number of observations per site
TOCS.1	OBS_TAB	SES	I4	3	4538	Observation tables: scan index, indices of the first and the second
TOCS.1	SITNAMES	SES	C1	8	10	IVS site names
TOCS.1	BAND_NAM	SES	C1	2	1	Band names
TOCS.1	CORPLACE	SES	C1	32	1	Correlator place name
TOCS.1	COR_TYPE	SES	C1	8	1	Correlator type: MK3 MK4 K4 S2 VLBA MITAKA-1
TOCS.1	COR_VERS	SES	C1	8	1	Correlator software and/or hardware version
TOCS.1	DEL_RATE	BAS	R8	2	1	Phase delay rates per band (d/l)
TOCS.1	EXP_CODE	SES	C1	32	1	Experiment code
TOCS.1	EXP_DESC	SES	C1	80	1	Experiment description
TOCS.1	EXP_NAME	SES	C1	80	1	Experiment program name
TOCS.1	GDAMBSP	BAS	R8	2	1	Group delay ambiguity spacings per band (sec)
TOCS.1	GRDELERR	BAS	R8	2	1	Group delay errors per band (sec)
TOCS.1	GRRATERR	BAS	R8	2	1	Group delay rate errors per band (d/l)
TOCS.1	GR_DELAY	BAS	R8	2	1	Group delays per band (sec)
TOCS.1	GR_RATE	BAS	R8	2	1	Group delays rate per band (d/l)
TOCS.1	ION_GDEL	BAS	R8	1	1	Ionospheric contribution to group delay at the first band (sec)
TOCS.1	ION_GERR	BAS	R8	1	1	Uncertainty of ionospheric contribution to group delay at the first band (sec)
TOCS.1	ION_PRAT	BAS	R8	1	1	Ionospheric contribution to phase delay rate at the first band (sec)
TOCS.1	ION_RERR	BAS	R8	1	1	Uncertainty of ionospheric contribution to phase delay rate at the first band (sec)
TOCS.1	MJD_OBS	SCA	I4	1	1	MJD of fringe reference time at pseudo-UTC timescale for the scan
TOCS.1	NUMB_SOU	SES	I4	1	1	Number of observed sources
TOCS.1	NUM_BAND	SES	I4	1	1	Number of frequency bands observed in the experiment
TOCS.1	NUM_CHAN	SES	I4	1	1	Number of frequency channels at all bands
TOCS.1	NUM_CHBN	SES	I4	1	2	Number of frequency channels per band
TOCS.1	N_AVBAND	SES	I4	1	1	Number of frequency bands for which information is available
TOCS.1	PHDELERR	BAS	R8	2	1	Phase delay error (rad)
TOCS.1	PHRATERR	BAS	R8	2	1	Phase delay rate delay errors per band (d/l)
TOCS.1	PI_NAME	SES	C1	80	1	Name of the principal investigator
TOCS.1	QUALCODE	BAS	C1	2	2	Quality code as char*2 value: 5-9 is good, 0 -- non-detection, left
TOCS.1	REC_MODE	SES	C1	80	1	Recording mode
TOCS.1	REF_FREQ	BAS	R8	2	1	Reference frequency for phase delay per band (Hz)
TOCS.1	SBDELERR	BAS	R8	2	1	Single-band delay errors per band (sec)
TOCS.1	SB_DELAY	BAS	R8	2	1	Single-band delays per band (sec)

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TOCS.1	SCANNAME	SCA	C1	16	1	Scan name
TOCS.1	SCANPIMA	SCA	C1	10	1	Pima internal scan name
TOCS.1	SIT_COOR	SES	R8	3	10	Site coordinates in a crust-fixed terrestrial reference system: X, Y, Z (m)
TOCS.1	SKYFRQCH	SES	R8	16	1	Sky frequency of channels in Hz
TOCS.1	SNRATIO	BAS	R8	2	1	Fringe amplitude signal to noise ratio (d/I)
TOCS.1	SOU_COOR	SES	R8	2	68	Source coordinates in a barycentric reference system: right ascension, declination, parallax (mas)
TOCS.1	SOU_IND	SCA	I4	1	1	Source name index
TOCS.1	SRCNAMES	SES	C1	8	68	Source names
TOCS.1	STA_IND	BAS	I4	2	1	Station names indexes
TOCS.1	TOTPHASE	BAS	R8	2	1	Total fringe phases at time of arrival signal at station 1 per band
TOCS.1	UTC_MTAI	SES	R8	1	1	Difference UTC minus TAI at first time tag of the database (sec)
TOCS.1	UTC_OBS	SCA	R8	1	1	Pseudo-UTC time tag of fringe reference time for the scan (sec)
TOCS.2	ANT_GAIN	STA	R4	2	1	Antenna gain per band K/Jy
TOCS.2	APRCLOOF	STA	R8	1	1	Apriori clock offset used for correlation
TOCS.2	APRCLORT	STA	R8	1	1	Apriori clock rate used for correlation
TOCS.2	APR_DEL	BAS	R8	2	1	Apriori delay (sec)
TOCS.2	APR_PHAS	BAS	R8	2	1	Apriori fringe phase (rad)
TOCS.2	APR_PHGC	BAS	R8	2	2	Apriori fringe phase per station at the geocenter (rad)
TOCS.2	APR_RATE	BAS	R8	2	1	Apriori delay rate (d/I)
TOCS.2	BITSAMPL	SES	I2	1	1	Number of bits per sample
TOCS.2	CHAN_SDB	SES	I2	16	1	Index of lower (-1) or upper (1) sideband per channel
TOCS.2	CHAN_WDT	SES	R8	16	1	Frequency channel width in Hz per channel
TOCS.2	DELW_CEN	BAS	R8	2	1	Delay window center used for fringe search (sec)
TOCS.2	DELW_WDT	BAS	R8	2	1	Delay window width used for fringe search (sec)
TOCS.2	FRN_AMPL	BAS	R8	2	1	Normalized fringe amplitude in range [0, 1]
TOCS.2	FRT_OFFS	SCA	R8	2	1	Fringe reference time offset relative to the scan start (sec)
TOCS.2	GCRESPHS	BAS	R8	2	1	Residual fringe phase at time of arrival signal at geocenter (rad)
TOCS.2	GC_PHASE	BAS	R8	2	1	Fringe phase at time of arrival signal at the conventional geocenter (rad)
TOCS.2	IND_CHN1	BAS	I2	10	1	Indexes of channels used in bandwidth synthesis in band 1
TOCS.2	IND_CHN2	BAS	I2	6	1	Indexes of channels used in bandwidth synthesis in band 2
TOCS.2	NOISERMS	BAS	R4	2	1	RMS of the fringe amplitude noise (d/I)
TOCS.2	NUM_AP1	BAS	I2	10	2	Number of accumulation periods used in band 1 per channel per scan
TOCS.2	NUM_AP2	BAS	I2	6	2	Number of accumulation periods used in band 2 per channel per scan
TOCS.2	NUM_SAM1	BAS	R8	10	1	Number of samples used in bandwidth synth. in band 1 per frequency
TOCS.2	NUM_SAM2	BAS	R8	6	2	Number of samples used in bandwidth synth. in band 2 per frequency
TOCS.2	NUSEDCHN	BAS	I2	2	1	Number of channels used in bandwidth synthesis per band
TOCS.2	PCAL_CM1	STA	R4	2	10	Complex phase cal (real/image) per channel at the 1st band
TOCS.2	PCAL_CM2	STA	R4	2	6	Complex phase cal (real/image) per channel at the 2nd band
TOCS.2	PCAL_FR1	STA	R8	1	10	Phase cal frequency per channel at the 1st band
TOCS.2	PCAL_FR2	STA	R8	1	6	Phase cal frequency per channel at the 2nd band
TOCS.2	POLARIZ	BAS	C1	2	2	Polarization label: RR, RL, LR, or LL
TOCS.2	RATE_CEN	BAS	R8	2	1	Rate window center used for fringe search (d/I)
TOCS.2	RATE_WDT	BAS	R8	2	1	Rate window width used for fringe search (d/I)
TOCS.2	RESGRRAT	BAS	R8	2	1	Residual group delay rate (d/I)
TOCS.2	RESMBDEL	BAS	R8	2	1	Residual multiband group delay (sec)
TOCS.2	RESPHAS	BAS	R8	2	1	Residual fringe phase
TOCS.2	RESPHRAT	BAS	R8	2	1	Residual phase delay rate (d/I)
TOCS.2	RESSBDEL	BAS	R8	2	1	Residual singleband group delay (sec)
TOCS.2	RES_GRDL	BAS	R8	2	1	Residual group delay reported by the post-correlator software (sec)
TOCS.2	RES_PHGC	BAS	R8	2	1	Residual fringe phase per station at the geocenter (rad)

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TOCS.2 RES_RATE BAS R8 2 1 Apriori phase delay rate reported by the post-correlator software
 TOCS.2 SAMPLRAT SES R8 1 1 Sample rate in Hz
 TOCS.2 SCAN_DUR BAS R8 2 1 Scan duration per band (sec)
 TOCS.2 SPCH_WDT SES R8 16 1 Spectral channel width in Hz per channel
 TOCS.2 TSYS1 STA R4 1 10 System temperature per channel at the 1st band (K)
 TOCS.2 TSYS2 STA R4 1 6 System temperature per channel at the 2nd band (K)
 TOCS.2 UVSTAORD BAS I2 1 1 Original station order in the baseline: 1 (ascending) or -1 (descen
 TOCS.2 UV_CHN1 BAS R4 2 10 UV data: real and image part per channel at the 1st band (d/I)
 TOCS.2 UV_CHN2 BAS R4 2 6 UV data: real and image part per channel at the 2nd band (d/I)
 TOCS.3 AIR_TEMP STA R8 1 1 Air temperature at the station (K)
 TOCS.3 ATM_PRES STA R8 1 1 Atmospheric pressure at the station (Pa)
 TOCS.3 CABL_DEL STA R8 1 1 Cable delay (sec)
 TOCS.3 CABL_SGN SES I2 10 1 Cable sign: +1 or -1
 TOCS.3 CAL_INFO SES I4 4 1 Information about class and type of available calibrations
 TOCS.3 CAL_NAME SES C1 8 1 Name of available calibrations
 TOCS.3 MEANCABL SES R8 10 1 Mean cable delay (sec)
 TOCS.3 N_CALIB SES I2 1 1 Number of available calibrations
 TOCS.3 REL_HUMD STA R8 1 1 Relative humidity at the station (0-1)

Specific LCODEs

The following LCODES hold information specific for a given data analysis software:

TOCS.1 EXPSERNO SES I2 1 1 Experiment serial number at correlator
 TOCS.1 MK3_DBNM SES C1 10 1 Mark-3 DBH database name
 TOCS.1 PIMA_CNT SES C1 128 1 Full path name of pima control file
 TOCS.1 PIMA_VER SES C1 24 1 Version of PIMA software
 TOCS.2 PIND_OBS BAS I4 1 1 Internal index of observation used by PIMA
 TOCS.2 SRT_OFFS SCA R8 1 1 Scan reference time offset relative to the scan start (sec)
 TOCS.4 ATM_CNS SES R8 10 32 Reciprocal weights of constraints on atm. path delay rate per sta
 TOCS.4 ATM_INTR SES R8 32 1 Length of time span between spline nodes for atm. path delay per
 TOCS.4 AUTO_SUP BAS I4 1 1 Bit field of automatic suppression status for combination of obser
 TOCS.4 BAND_2ND BAS I2 1 1 Bit field with status of information about the second band observ
 TOCS.4 BAS_USE SES I4 45 1 Bit field of baseline selection status
 TOCS.4 BSCL_CNS SES R8 45 32 Reciprocal weights of constraints on basdep. clock, per baseline
 TOCS.4 BSCL_EST SES I4 45 1 Estimation status for baseline dependent clock, per baseline
 TOCS.4 CAL_STS SES I4 10 1 Bit field of using available calibrations per station, per calibration
 TOCS.4 CLO_CNS SES R8 10 32 Reciprocal weights of constraints on clock rate per station, per s
 TOCS.4 CLO_INTR SES R8 32 1 Length of time span between spline nodes for clock function per
 TOCS.4 DATYP SES I2 1 1 Type of the observable or a combination of observables used in the
 TOCS.4 DGCL_EST SES I2 10 32 Degree of global clock function polynomial per station, per soluti
 TOCS.4 EDIT_STS SES I4 1 1 Bit field of database editing status for different types of solutions
 TOCS.4 EFF_FREQ BAS R8 3 2 Effective ionospheric frequencies for gr.del, ph.del, ph.rate per b
 TOCS.4 EOP_CNS SES R8 11 32 Reciprocal weights of constraints on EOP related parameters
 TOCS.4 EOP_EST SES I4 11 1 Estimation status for EOP-related parameters
 TOCS.4 NUM_CLBR SES I4 1 1 Number of clock breaks in the experiment
 TOCS.4 NUM_CLRF SES I4 1 1 Number of clock reference stations
 TOCS.4 N_GRAMB BAS I4 2 1 Number of group delay ambiguities to be added to measured grou
 TOCS.4 N_PHAMB BAS I4 2 1 Number of phase delay ambiguities to be added to measured phas
 TOCS.4 RWBASNAM SES C1 16 45 Baseline names for additive baseline-dependent reweighting p

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TOCS.4 RWDELVAL	SES	R8	32	45	Additive baseline-dependent reweighting parameters for delays
TOCS.4 RWRATVAL	SES	R8	32	45	Additive baseline-dependent reweighting parameters for delay r
TOCS.4 SOCO_CNS	SES	R8	68	32	Reciprocal weights of constraints on source coordinates per ob
TOCS.4 SOCO_EST	SES	I4	2	68	Estimation status for source coordinats per component, per obje
TOCS.4 SOU_USE	SES	I4	68	1	Bit field of source selection status
TOCS.4 STA_CLRF	SES	C1	8	1	Names of clock reference stations
TOCS.4 STA_USE	SES	I4	10	1	Bit field of station selection status
TOCS.4 STPS_CNS	SES	R8	10	32	Reciprocal weights of constraints on site positions per site, per
TOCS.4 STPS_EST	SES	I4	3	10	Estimation status for station positions per component, per station
TOCS.4 SUPMET	SES	I2	1	1	Code of the suppression method used in the solution
TOCS.4 TEC_STS	SES	I4	10	1	Flag of availability/usage of the external ionosphere calibration
TOCS.4 TIL_INTR	SES	R8	32	1	Lenght of time span between spline nodes for atmospphere tilt per
TOCS.4 TLOF_CNS	SES	R8	10	32	Reciprocal weights of constraints on atm. tilt offset per station,
TOCS.4 TLRT_CNS	SES	R8	10	32	Reciprocal weights of constraints on atm. tilt rate per station, pe
TOCS.4 USER_REC	BAS	I4	1	1	Bit field of analyst defined recovery status for combination of obs
TOCS.4 USER_SUP	BAS	I4	1	1	Bit field of analyst defined suppression status for combination of
TOCS.5 APR_EOP	BAS	R8	3	2	Aprori EOP array as Euler angles and its derivatives (rad)
TOCS.5 AZIMUTH	STA	R8	1	1	Apparent source azimuth at both stations of the baseline (rad)
TOCS.5 DER_DEL	BAS	R8	64	1	Array of partial derivatives of theoretical path delay wrt paramete
TOCS.5 DER_RAT	BAS	R8	64	1	Array of partial derivatives of theoretical delay rate wrt parameter
TOCS.5 ELEV	STA	R8	1	1	Apparent source elevation at both stations of the baseline (rad)
TOCS.5 EOP_TAB	SES	R8	15	3	Table of aprori EOP as Euler angles with frequencies > 2 cpd filte
TOCS.5 MJD_EOP	SES	I4	1	1	Modified Julian date of the first epoch for the table of apriori EOP
TOCS.5 NUT_DER	BAS	R8	2	1	Partial derivatives wth nutation deaily offset parameters (sec)
TOCS.5 N_APREOP	SES	I4	1	1	Number of nodes with apriori EOP
TOCS.5 STEP_EOP	SES	I4	1	1	Step of the EOP table of apriori EOP (sec)
TOCS.5 TAI_EOP	SES	I4	1	1	TAI time tag of first epoch of the table of apriori EOP (sec)
TOCS.5 THGR_DEL	BAS	R8	1	1	Theoretical group delay (sec)
TOCS.5 THPH_DEL	BAS	R8	1	1	Theoretical phase delay (sec)
TOCS.5 THPH_RAT	BAS	R8	1	1	Theoretical phase delay date (d/l)
TOCS.5 TH_PROG	SES	C1	64	1	Name and version of the program which computed theoretical pa
TOCS.5 TH_RUDAT	SES	C1	24	1	Date and time of theoretical delay compuaton
TOCS.5 UV_COOR	BAS	R8	2	1	UV coordinates of the baseline vector projectioncalibration

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